# 141. Adolf Baeyer and Victor Villiger: On the Action of Caro's Reagent on Ketones.

*[Third, preliminary communication from the Chemical Laboratory of the Academy of Sciences at Munich.]*

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*English translation of Ber. dtsch. chem. Ges. 33 (1900) 858–864. Structural formulas are the originals. Article No. 142 (Tambor), which begins on the final page, is a separate paper, not included.*

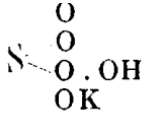
In the two earlier communications[1]) we described the action of Caro's reagent on ketones, and now add some closer details about the bodies thereby formed, prefacing them with a remark on the constitution of Caro's reagent.

## On the Nature of Caro's Reagent

If one dissolves benzoyl peroxide in ether and adds to the solution the amount of sodium in alcohol corresponding to one atom, a white, powdery precipitate of a sodium compound separates, while the ethyl ester of benzoic acid remains in solution. This sodium compound behaves like the sodium salt of an extremely weak acid, one which is already precipitated as a crystalline powder from the aqueous solution by carbonic acid. The process is evidently the following:

$$C_6H_5CO.O.O.COC_6H_5 + NaOC_2H_5 = C_6H_5CO.O.ONa$$
$$- C_6H_5.COOC_2H_5.$$

This hitherto unknown monobenzoyl hydrogen peroxide separates black iodine instantly from potassium iodide solution and gives crystalline nitrosobenzene with aniline water, thus behaving exactly like Caro's reagent. Acetyl peroxide shows a similar behavior. The assumption we made in the second communication — that Caro's reagent has a composition analogous to the lead-chamber crystals and corresponds to the formula



— hereby receives an experimental confirmation. More precise data on the properties of benzoyl hydrogen peroxide we shall give shortly.

### Acetone

Wolffenstein[2]) prepared, by prolonged action of acetone on concentrated hydrogen peroxide, an acetone peroxide of m.p. 97°; we obtained, according to the first communication, a peroxide melting at 132–133° by the action of Caro's reagent on acetone. This latter reaction occurs instantaneously; but one can also obtain Wolffenstein's peroxide instantly if, to a mixture of about equal weights of ca. 50-percent hydrogen peroxide and acetone, concentrated hydrochloric acid is added under good cooling. The liquid solidifies to a paste of crystals which crystallize in centimeter-sized flat rhombohedra, presumably containing crystal alcohol, since they weather in air. To free them of it they were steam-distilled and then showed m.p. 90–94°. If, instead of hydrochloric acid, one uses sulfuric acid monohydrate (sulfuric acid treated with one molecule of water), the liquid likewise solidifies to a paste of crystals which for the most part melt indistinctly at 90°, while the last crystal particles disappear only at 105°. There thus probably forms here a mixture of Wolffenstein's and of our acetone peroxide.

The peroxide prepared with Caro's reagent, melting at 132–133°, gave on molecular-weight determination by the freezing-point method results showing it to be dimolecular. To see whether the peroxide prepared with hydrochloric acid, m.p. 90–94°, is identical with Wolffenstein's, a molecular-weight determination was made of it as well, which led to numbers agreeing with a trimolecular peroxide. This body is therefore identical with Wolffenstein's.

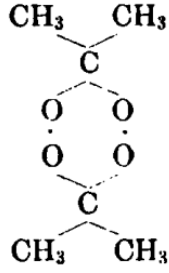
Determination of the constants for the benzene used, with the aid of naphthalene (m = 128).

| Benzene | Naphthalene | p | c | m · c / p | Constant (mean) |
|---|---|---|---|---|---|
| 29.70 g | 0.4550 g | 1.532 | 0.641° | 53.6 | 52.7 |
| | 0.9190 g | 3.094 | 1.252° | 51.8 | |

Molecular-weight determination of the acetone peroxides.

| | Benzene | Substance | p | c | m = 52.7 · p / c | m (calc.) |
|---|---|---|---|---|---|---|
| Peroxide, m.p. 132–133° | 29.75 g | 0.1951 g | 0.656 | 0.237° | 145.9 | $(C_3H_6O_2)_2$ = 148 |
| | | 0.5051 g | 1.698 | 0.604° | 148.1 | |
| | | 0.9072 g | 3.049 | 1.072° | 149.9 | |
| Peroxide, m.p. 90–94° | 19.09 g | 0.1975 g | 1.035 | 0.248° | 219.9 | $(C_3H_6O_2)_3$ = 222 |
| | | 0.5608 g | 2.938 | 0.695° | 222.8 | |

To the peroxide of m.p. 132–133° there accordingly probably belongs the formula



and it contains a six-membered ring, whereas a nine-membered ring should underlie Wolffenstein's peroxide.

A chlorinated, polymeric acetone peroxide is obtained, according to preliminary experiments, when chloroacetone is treated under ice cooling with a mixture of potassium persulfate and sulfuric acid monohydrate. The product consists of an oil, not examined more closely, and of a solid body which crystallizes from methyl alcohol in oblique prisms of m.p. 104–105°, explodes on impact, and is without action on potassium iodide solution.

## Menthone

It was already stated in the first communication that menthone, treated with the dry Caro reagent, yields a very good amount of lactone. This can be increased still further if the dry reagent is poured over with glacial acetic acid before the addition of the menthone. 1 g of menthone thus yielded 0.9 g of crude, crystallized lactone.

### Ethyl ester of 2,6-dimethyloctan-3-ol-8-oic acid

If one lets menthone lactone stand with alcoholic sulfuric acid for 24 hours, it is quantitatively converted into the ethyl ester of the corresponding hydroxy acid. This cleavage of the lactone likewise occurs when the menthone is brought together, in alcoholic solution, with the liquid reagent. The yield of the ester is very considerable.

5 g of menthone lactone were left standing for 24 hours with 50 g of sulfuric acid monohydrate and 35 g of absolute alcohol. After appropriate treatment, 5.6 g of an oil boiling at 152–155° under 15 mm pressure were obtained, of the composition $C_{10}H_{19}O_2.OC_2H_5$, which is the ethyl ester of 2,6-dimethyloctan-3-ol-8-oic acid.

0.1713 g subst.: 0.4206 g $CO_2$, 0.1712 g $H_2O$.

$C_{12}H_{24}O_3$ Calc. C 66.67, H 11.11. Found C 66.96, H 11.10.

36 g of potassium persulfate were treated with 110 g of sulfuric acid monohydrate, then, under cooling, 70 g of absolute alcohol were added and 10 g of menthone dripped in within a quarter of an hour under ice cooling and stirring. After standing one hour in ice, the product was poured onto ice, extracted with ether, and the oil obtained after washing with water and caustic soda — which, owing to the presence of some peroxide, reacted on potassium iodide — was washed with bisulfite and then fractionated in vacuum. Besides a small menthone-containing forerun, 8.7 g of an oil boiling at 136–155° under 15 mm were obtained. On repeated fractionation the liquid passing over at about 150° proved identical with the ethyl ester described above, gave on analysis nearly corresponding numbers, and on saponification the crystallizing hydroxy acid melting at 66.5°. Analysis of the ethyl ester:

0.1515 g subst.: 0.3730 g $CO_2$, 0.1497 g $H_2O$.

$C_{12}H_{24}O_3$ Calc. C 66.67, H 11.11. Found C 67.15, H 10.98.

### Methylcyclohexanone

The methylcyclohexanone was prepared from pulegone by Wallach's method as modified by Klages[1]). In the attempt to

prepare the lactone by the method given for menthone, with the dry reagent and glacial acetic acid, only a simple peroxide acting strongly on potassium iodide was obtained. The lactone could not be obtained in any tolerably satisfactory way by other means either, but the ester of the corresponding hydroxy acid could.

### Ethyl ester of methylhexanolic acid

From 10 g of methylcyclohexanone, according to the procedure given for menthone, 7.7 g of an oil of b.p. 140–146° at 15 mm were obtained. After renewed distillation the main fraction, 141–142.5° at 15 mm, was analyzed.

0.1500 g subst.: 0.3432 g $CO_2$, 0.1373 g $H_2O$.

$C_9H_{18}O_3$ Calc. C 62.07, H 10.34. Found C 62.40, H 10.17.

### Methylhexanolic acid

On saponification of the ethyl ester with dilute aqueous caustic soda, a small amount of an indifferent, crystallizing substance remained behind. The acid is a syrup that dissolves readily in water and, with the exception of the silver salt, yields no characteristic salts. The latter crystallizes on cooling from a hot solution in little needles; its analysis agreed with the composition of the silver salt of a methylhexanolic acid.

0.2143 g subst.: 0.2606 g $CO_2$, 0.0996 g $H_2O$. — 0.2332 g subst.: 0.0991 g Ag.

$C_7H_{13}O_3Ag$ Calc. C 33.20, H 5.14, Ag 42.69. Found C 33.17, H 5.16, Ag 42.50.

Depending on whether the lactone formation occurs on the one or the other side of the carbonyl, the acid must be either a 3-methylhexan-6-ol-1-oic acid or a 2-methylhexan-1-ol-6-oic acid. This question could not yet be decided, owing to the difficulty of investigating the methyladipic acid prepared from the hydroxy acid.

### Suberone

The lactone derived from suberone is of particular interest, because it belongs to the still-unknown class of the ζ-lactones. Unfortunately it has not so far been possible for us to prepare this

lactone in the pure state, since a larger quantity of suberone than was at our disposal would be required for this.

In general there arise, on the action of Caro's reagent on ketones, three classes of substances: simple peroxides, polymerized peroxides, and lactones. These bodies now form from suberone too, in varying amount according to the conditions.

When, for example, a solution of 1 g of suberone in 15 g of glacial acetic acid was dripped under cooling into the dry reagent prepared from 7 g of persulfate, there were obtained, after two hours' standing in ice, 0.15 g of a neutral oil, consisting for the most part of the lactone, since on saponification it yielded a copious amount of a hydroxy acid which, on warming with hydriodic acid, passed into the iodoönanthic acid described below.

Much better, on the other hand, is the preparation of the ethyl ester of oxyönanthylic acid. Since the formation of the latter, as shown above, is in all probability preceded by the formation of the lactone, it is to be assumed that the yield of the latter is so small only because of its ready decomposability.

To a cooled mixture of 50 g of potassium persulfate with 150 g of sulfuric acid monohydrate and 100 g of alcohol, 10 g of suberone were dripped within a quarter of an hour. After two hours' standing in ice, the mixture was diluted to 1 L by addition of ice and water. A considerable amount of crystals thereby separated, which were filtered off.

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[1]) The two earlier communications: Diese Berichte 32, 3625; 33, 124. [2]) Diese Berichte 28, 2265. (methylcyclohexanone) [1]) Diese Berichte 32, 2567.

### Polymeric suberone peroxide

These crystals are a polymeric modification of suberone peroxide. The yield amounted to 2.1 g; purification was effected by recrystallization from ether, from which the substance crystallizes in rhombic leaves of m.p. 99–100°.

0.2104 g subst.: 0.5086 g $CO_2$, 0.1767 g $H_2O$.

$C_7H_{12}O_2$ Calc. C 65.63, H 9.37. Found C 65.93, H 9.33.

The substance is sparingly soluble in alcohol and ether, readily soluble in benzene and chloroform. It is volatile with water vapor, distils undecomposed on heating in a test tube, and deflagrates in

contact with concentrated sulfuric acid. Toward potassium iodide it is stable, likewise toward boiling caustic soda.

### Ethyl ester of ζ-oxyönanthylic acid

From the clear filtrate left by the peroxide, the ethyl ester of ζ-oxyönanthylic acid was isolated by extraction with ether and purified by distillation in vacuum. The boiling point was found at 146–155° under 15 mm pressure. The yield amounted to 6.2 g.

0.1424 g subst.: 0.3239 g $CO_2$, 0.1302 g $H_2O$.

$C_9H_{18}O_3$ Calc. C 62.07, H 10.34. Found C 62.03, H 10.16.

The ester is an oil fairly soluble in water and is readily saponified by caustic soda.

### ζ-Oxyönanthylic acid (heptan-7-ol-1-oic acid)

The hydroxy acid is miscible with water in all proportions and is precipitated as a syrup from the solution by ammonium sulfate. The calcium salt is readily soluble and crystallizes from the concentrated solution. The zinc salt is sparingly soluble and crystallizes in small prisms (characteristic). The lead salt forms sparingly soluble needles; the silver salt, a fairly sparingly soluble crystalline precipitate.

On analysis of the silver salt, somewhat too little carbon and hydrogen and too much silver were found. This probably arises from an admixed trace of pimelic acid resulting from an oxidation carried too far.

0.2240 g subst.: 0.2656 g $CO_2$, 0.0991 g $H_2O$. — 0.2045 g subst.: 0.0888 g Ag.

$C_7H_{13}AgO_3$ Calc. C 33.20, H 5.14, Ag 42.69. Found C 32.34, H 4.92, Ag 43.42.

### 7-Iodoheptan-1-oic acid

If one boils the hydroxy acid for 10 minutes with distilled hydriodic acid, the iodoheptanoic acid separates in crystallizing oil drops. For purification the acid was repeatedly recrystallized from light-boiling benzine and then formed lustrous leaves of m.p. 49–51°. It is readily soluble in all solvents with the exception of water and cold benzine. The corresponding ζ-bromoönanthylic acid,

obtained analogously, is still more readily soluble and melts at 30–31°.

0.2644 g subst.: 0.3171 g $CO_2$, 0.1242 g $H_2O$. — 0.2159 g subst.: 0.1966 g AgI.

$C_7H_{13}IO_2$ Calc. C 32.81, H 5.08, I 49.61. Found C 32.71, H 5.22, I 49.21.

To convert the oxyönanthylic acid into pimelic acid, the ester was saponified with caustic soda, carbon dioxide was passed in, and the solution was heated for 5 minutes on the water bath with an excess of permanganate solution. The pimelic acid obtained crystallized from hot water in leaves of m.p. 103–105°. It was identified by comparison of its calcium and silver salts with those of normal pimelic acid.